

Modern Tools for Arithmetical Completeness

Provability Models and Proofs

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01 Provability Logic

Language

We will work with the usual language of classical modal logic:

$$\phi ::= p \mid \perp \mid \phi \rightarrow \phi \mid \Box \phi.$$

As usual, we define the rest of connectives

$$\neg \phi := \phi \rightarrow \perp,$$

$$\phi \vee \psi := \neg \phi \rightarrow \psi,$$

$$\phi \wedge \psi := \neg(\neg \phi \vee \neg \psi),$$

$$\Diamond \phi := \neg \Box \neg \phi.$$

The idea is that we want to interpret $\Box \phi$ as ‘ ϕ is provable in some mathematical (arithmetical) theory T ’.

Why provability logic?

1. At the origin of modal logic (Orloff; 1928), (Gödel; 1933).
2. Provides an analysis of second Gödel incompleteness: arithmetical content vs modal content.
3. Provides an axiomatic approach to the study of the notion of provability (Beklemishev and Visser; 2006).
4. Can be used for classical proof theory, e.g. for ordinal analysis (Beklemishev; 2004).

Arithmetical semantics

Let us fix PA as the arithmetic theory of interest, and let $\ulcorner \cdot \urcorner$ be an ‘adequate’ Gödel encoding (satisfy diagonal lemma). Given an arithmetic interpretation $(\cdot)^* : \text{PropVar} \longrightarrow \text{ArithSent}$ we extend it to all modal formulas as follows:

$$\perp^* := \perp, \quad (\phi \rightarrow \psi)^* := \phi^* \rightarrow \psi^*, \quad (\Box \phi)^* := \text{Prv}_{\text{PA}}(\ulcorner \phi^* \urcorner).$$

Then we are interested in the following set of formulas

$$\mathcal{PL}_{\text{PA}}(\text{PA}) = \{\phi \mid \text{PA} \vdash \phi^* \text{ for any } (\cdot)^*\}.$$

More generally, $\mathcal{PL}_T(U) = \{\phi \mid T \vdash \phi^* \text{ for any } (\cdot)^*\}$ where $\Box$ is interpreted as Prov_U .

The logic GL

We define the logic GL as the smallest set of formulas containing:

1. Propositional tautologies.
2. $\Box(\phi \rightarrow \psi) \rightarrow \Box\phi \rightarrow \Box\psi$ for any formulas ϕ, ψ .
3. $\Box(\Box\phi \rightarrow \phi) \rightarrow \Box\phi$ for any formula ϕ .

and is closed under the rules

$$\frac{\phi \quad \phi \rightarrow \psi}{\psi} \text{MP}$$

$$\frac{\phi}{\Box\phi} \Box^{\text{GL}}$$

Lemma

We have that $\text{GL} \vdash \Box\phi \rightarrow \Box\Box\phi$.

Kripke semantics

A *transitive conversely well-founded Kripke frame* is an ordered pair $\mathcal{M} = (W, R)$ where

- W is a non-empty set.
- R is a binary relation on W that is *transitive* and *conversely well-founded*. In words, there is no sequence $(w_i)_{i \in \mathbb{N}}$ such that

$$w_0 R w_1 R w_2 R \cdots R w_n R w_{n+1} R \cdots .$$

A GL-*Kripke model* is a triple $\mathfrak{M} = (W, R, V)$ where (W, R) is a Kripke frame and $V : W \rightarrow \wp(\text{Prop})$. This function is called a *valuation*.

Validity on Kripke frames

We remember the relation of satisfaction for Kripke models.

Definition

Let $\mathfrak{M}$ be a Kripke model, w a world and ϕ a formula. We define $\mathfrak{M}, w \Vdash \phi$ as

- $\mathfrak{M}, w \Vdash p$ iff $p \in V(w)$.
- $\mathfrak{M}, w \Vdash \perp$ iff never.
- $\mathfrak{M}, w \Vdash \phi \rightarrow \psi$ iff $\mathfrak{M}, w \nVdash \phi$ or $\mathfrak{M}, w \Vdash \psi$.
- $\mathfrak{M}, w \Vdash \Box \phi$ iff for any $v \in R[w]$, $\mathfrak{M}, v \Vdash \phi$.

For a frame $\mathcal{M}$ we write $\mathcal{M} \models \phi$ if for any valuation V and any world w we have that $(\mathcal{M}, V), w \Vdash \phi$.

Kripke Soundness & Completeness

The important result about Kripke semantics is the following.

Theorem

$GL = \{\phi \mid \mathcal{F} \models \phi, \text{ for any transitive cwf frame } \mathcal{F}\}.$

$\subseteq$ is called soundness and $\supseteq$ is called completeness.

In fact we can add *finite* to the class of frames and *rooted tree*.

Arithmetic Soundness & Completeness

Then (Solovay; 1976) showed the following.

Theorem

$GL = \{\phi \mid PA \vdash (\phi)^*, \text{ for any arith. interp. } (\cdot)^*\}.$

Soundness is a proof by induction on the length of the Hilbert proof (and was known before Solovay, (Löb; 1955)).

Let us sketch how Solovay proved completeness.

Sketch of Solovay's proof

Solovay's proof can be divided in two steps.

1. Show that GL is complete for a simple class of Kripke frames (the finite irreflexive transitive rooted trees).
2. Assume that $GL \not\vdash \phi$, obtain a countermodel of ϕ which is a finite irreflexive transitive rooted tree. By recursion (and exploiting the simplicity of the model), embed of the countermodel inside arithmetic (defining an arithmetical interpretation).

Quick view of the literature

Same procedure, but Kripke frames tend to not work.

- For ER Carlson (Carlson; 1986) uses the class of Kripke frames for a sublogic CSM that satisfy the missing axiom. Visser (Visser; 1995) uses topological general Kripke semantics and modifies Solovay's method to embed this models in arithmetic.
- For GLP (Japaridze; 1986) used a complete but unsound semantics while (Ignatiev; 1993) and (Beklemishev; 2011) use a Kripke complete sublogic.

“Quick” view of the literature

- Interpretability: Friedmann models (Visser; 1990), simplified models (Shavrukov; 1988), (Berarducci; 1990) and Veltman models (Zambella; 1992).
- Non-normal modal logics: (Kogure and Kurahashi I; 2026) uses N-models, (Kogure; 2025) uses neighbourhood semantics and (Kogure and Kurahashi II; 2026) uses a new semantics (close to provability models).
- Intuitionistic Provability Logic: (Arideshir and Mojtahedi; 2017), (Mojtahedi; 2022) use **provability models** to reduce the provability logic to the Σ_1 -provability logic.

Two Questions

Then, two (related but not equal) questions arise.

Q1 What should be the semantics of provability logics?

Q2 Can we find a more uniform approach to the first step of Solovay's proof?

The rest of the talk is divided in two parts:

1. In sections 02 and 03 we will tackle **Q1** with **provability models**.
2. In section 04 we will propose a solution to **Q2** with **(sequent) proofs**.

02

Provability Models

j.w.w. Mojtaba Mojtahedi

Motivations of Provability Models

1. They arise naturally in the current characterization of the provability logic of HA.
2. Having a natural semantics for provability logics which are Kripke incomplete.
3. Provide a flexible interchange between 'structural' properties and 'syntactic' properties. More of this later.

Logic

A *logic* L is a pair $(Ax(L), Rul(L))$ such that

1. $Ax(L)$ is a set of formulas.
2. $Rul(L)$ is a set of rules, where rules are just a set of pairs of a set of formulas and a formula.

Some rules are

$$\frac{\phi}{\psi} \quad \frac{\phi \rightarrow \psi}{\psi} \text{ mp} \quad \frac{\phi}{\Box \phi} \text{ nec} \quad \frac{\Box \phi \rightarrow \phi}{\phi} \text{ Löb}$$

We define $\Gamma \vdash_L \phi$ as usual (sequence obtained from axioms and formulas in Γ applying rules on previous formulas).

Note: logics do not have to be closed under substitution.

Local vs Global

Note that we could define GL in two ways:

1. GL^ℓ has inference rule mp and axioms

$$\Box \text{PropTaut} \cup \Box \{ \Box \phi \rightarrow \Box \Box \phi \mid \phi \in \mathcal{L}_\Box \} \cup$$

$$\Box \{ \Box (\Box \phi \rightarrow \phi) \rightarrow \Box \phi \mid \phi \in \mathcal{L}_\Box \}$$

2. GL^g has inferences rules mp, nec and axioms

$$\text{PropTaut} \cup \{ \Box (\Box \phi \rightarrow \phi) \rightarrow \Box \phi \mid \phi \in \mathcal{L}_\Box \}$$

We will take GL^ℓ to be the adequate one (it corresponds with local semantics consequence in Kripke models).

A logic is *classic* if it contains all classical tautologies (in the language $\mathcal{L}_\Box$) and modus ponens as one of its rules. Both logics above are classical.

Accessibility relation

Let $\sqsubset$ be a binary relation. We define

1. $\sqsubset$ is a tree if $w, v \sqsubset u$ implies $w \sqsubseteq^+ v$ or $v \sqsubseteq^+ w$;
2. $w \sqsubset^1 v$ as the immediate predecessor, i.e., $w \sqsubset v$ and there is no u such that $w \sqsubset^+ u \sqsubset^+ v$;
3. $w \sim_{\sqsubset} v$ if $w = v$ or there is some u with $u \sqsubset^1 w$ and $u \sqsubset^1 v$;
4. $w \hat{\sqsubset} v$ if there is a u such that $w \sim_{\sqsubset} u \sqsubset^+ v$.

Lemma

If $(W, \sqsubset)$ is a conversely well-founded tree, then $\hat{\sqsubset}$ is conversely well-founded.

Provability pre-models

Definition

A *pre-model* is a tuple $\mathcal{P} = (W, \sqsubset, \{L_w\}_{w \in W}, V)$ such that

1. $\mathcal{K}_{\mathcal{P}} = (W, \sqsubset, V)$ is a Kripke model, and
2. for any $w \in W^{\sqsubset}$, L_w is a *classical* logic,

where $W^{\sqsubset} = \{v \in W \mid \exists w \in W. w \sqsubset v\}$.

The relation of satisfiability in a world is defined as in Kripke models, with the exception of

$$\mathcal{P}, w \models \Box \phi \quad \text{iff} \quad \forall v \sqsupset w. \vdash_v \phi.$$

Provability models

Additionally, we define

$$\mathcal{P}, w \models^+ \phi \quad \text{iff} \quad \exists v \sqsubset w \forall u \sqsupset^+ v. \mathcal{P}, u \models \phi.$$

A formula is said to be *purely modal* if it is a boolean combination of $\Box$ -formulas.

Definition (Modal Completeness)

A *provability model* is a provability pre-model $\mathcal{P}$ such that for every purely modal formula ϕ , if $\mathcal{P}, w \models^+ \phi$ then $\vdash_w \phi$.

From Kripke

Lemma

For every Kripke model $\mathcal{K} = (W, \sqsubseteq, V)$ there is a provability model $\mathcal{P} = (W, \sqsubseteq, \{L_w\}_{w \in W}, V)$ such that for any formula ϕ and world w

$$\mathcal{K}, w \models \phi \quad \text{iff} \quad \mathcal{P}, w \models \phi.$$

If $\mathcal{K}$ is transitive, we can make every L_w with necessitation.

From Kripke (Non-Transitive)

Define $\text{Ax}(\mathcal{L}_w) = \{\phi \mid \mathcal{K}, w \models \phi\}$ and $\text{Rul}(\mathcal{L}_w) = \{\text{mp}\}$. We prove the equivalence by induction in ϕ , the only interesting case is if $\phi = \Box \phi_0$.

If $\mathcal{K}, w \models \Box \phi_0$ then for every $v \sqsubset w$ we have that $\mathcal{K}, v \models \phi_0$ so $\phi_0 \in \text{Ax}(\mathcal{L}_v)$. Then $\vdash_v \phi_0$ for $v \sqsubset w$ so $\mathcal{P}, w \models \Box \phi_0$

Now, assume $\mathcal{P}, w \models \Box \phi_0$ so for any $v \sqsubset w$ we have that $\vdash_v \phi_0$. Note that, by definition, $\text{Ax}(\mathcal{L}_v)$ is closed under mp, so $\vdash_v \phi_0$ implies $\phi_0 \in \text{Ax}(\mathcal{L}_v)$. This implies that $\mathcal{K}, v \models \phi_0$ for $v \sqsubset w$, so $\mathcal{K}, w \models \Box \phi_0$, as desired.

Once the equivalence has been established, modal completeness is trivial.

From Kripke (Transitive)

Define $Ax(L_w) = \{\phi \mid \mathcal{K}, w \models \Box \phi\}$ and $Rul(L_w) = \{mp, nec\}$. We prove the equivalence by induction in ϕ , the only interesting case is if $\phi = \Box \phi_0$.

If $\mathcal{K}, w \models \Box \phi_0$ then for every $v \sqsubset w$ we have that $\mathcal{K}, v \models \Box \phi_0$ (by transitivity) so $\phi_0 \in Ax(L_v)$. Then $\vdash_v \phi_0$ for $v \sqsubset w$ so $\mathcal{P}, w \models \Box \phi_0$.

Now, assume $\mathcal{P}, w \models \Box \phi_0$ so for any $v \sqsubset w$ we have that $\vdash_v \phi_0$. Note that, by definition, $Ax(L_v)$ is closed under mp and nec , so $\vdash_v \phi_0$ implies $\phi_0 \in Ax(L_v)$. This implies that $\mathcal{K}, v \models \phi_0$ for $v \sqsubset w$, so $\mathcal{K}, w \models \Box \phi_0$, as desired.

Once the equivalence has been established, modal completeness is trivial.

Future work

1. Can we find an alternative condition to modal completeness?
2. Are there other natural ways of mixing syntax and semantics?
3. Can we find the notion of provability frame?

03

The case of GL

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Provability models of GL

The important point of provability models is the interchangeability of conditions on $\Box$ and of conditions on $(L_w)_{w \in W}$. For this reason, a GL provability model is a provability model such that

1. $\Box$ is transitive or each L_w has necessitation,
2. $\Box$ is conversely well-founded or each L_w has Löb's rule.

We proceed to show soundness and completeness for this class of models.

Soundness: propositional tautologies and axiom (K)

If ϕ is a propositional tautology it is straightforward to show that $\mathcal{P}, w \models \phi$. For $\Box \phi$ it follows from the fact that each logic has classical tautologies.

To show that $\mathcal{P}, w \models \Box(\phi \rightarrow \psi) \rightarrow \Box \phi \rightarrow \Box \psi$ it follows from each L_v being closed under mp. Now, let us show that $\mathcal{P}, w \models \Box(\Box(\phi \rightarrow \psi) \rightarrow \Box \phi \rightarrow \Box \psi)$. It suffices to show that for any world $v \in W^\square$ we have $\vdash_v \Box(\phi \rightarrow \psi) \rightarrow \Box \phi \rightarrow \Box \psi$. Note that we already know that $\mathcal{P} \models \Box(\phi \rightarrow \psi) \rightarrow \Box \phi \rightarrow \Box \psi$, i.e., for every $v \in W^\square$ $\mathcal{P}, v \models^+ \Box(\phi \rightarrow \psi) \rightarrow \Box \phi \rightarrow \Box \psi$. By modal completeness we conclude the desired result.

Soundness: axiom (4)

Assume all the logics of $\mathcal{P}$ have necessitation. Then $\mathcal{P}, w \models \Box \phi \rightarrow \Box \Box \phi$ follows from this, and $\mathcal{P}, w \models \Box(\Box \phi \rightarrow \Box \Box \phi)$ follows using modal completeness and the previous fact.

Now, let $\sqsubset$ be transitive and let us show $\mathcal{P}, w \models \Box \phi \rightarrow \Box \Box \phi$. Assume $\mathcal{P}, w \models \Box \phi$ and let $v \sqsubset w$. We need to show that $\vdash_v \Box \phi$, by modal completeness it suffices to show that $\mathcal{P}, v \models^+ \Box \phi$. For this, we are going to show that for any $u \sqsubset^+ w$ we have that $\mathcal{P}, u \models \Box \phi$. So let $w \sqsubset^+ u \sqsubset u'$ so, by transitivity, $w \sqsubset u'$ and then $\vdash_{u'} \phi$, as desired. That $\mathcal{P}, w \models \Box(\Box \phi \rightarrow \Box \Box \phi)$ follows from modal completeness and the previous fact.

Soundness: Löb's axiom

Assume each $(L_w)_{w \in W^\square}$ has Löb's rule, then $\mathcal{P}, w \models \square(\square\phi \rightarrow \phi) \rightarrow \square\phi$ trivially. Assume $\square$ is conversely well-founded, we proceed by induction on this order. So assume that for any $v \sqsupset^+ w$ we have that $\mathcal{P}, v \models \square(\square\phi \rightarrow \phi) \rightarrow \square\phi$. Then for any $v \sqsupset w$ we have that $\mathcal{P}, v \models^+ \square(\square\phi \rightarrow \phi) \rightarrow \square\phi$. By modal completeness $\vdash_v \square(\square\phi \rightarrow \phi) \rightarrow \square\phi$ for $v \sqsupset w$. Assume that $\mathcal{P}, w \models \square(\square\phi \rightarrow \phi)$, note that then (since (4) is globally valid) we have that $\mathcal{P}, w \models \Box(\square\phi \rightarrow \phi)$. Let $u \sqsupset w$, then $\vdash_u (\square\phi \rightarrow \phi)$ and $\vdash_u \square(\square\phi \rightarrow \phi)$. Since $\vdash_u \square(\square\phi \rightarrow \phi) \rightarrow \square\phi$ we obtain by mp that $\vdash_u \square\phi$. Using mp again and that $\vdash_u \square\phi \rightarrow \phi$ we get the desired $\vdash_u \phi$. The boxed axiom follows from modal completeness.

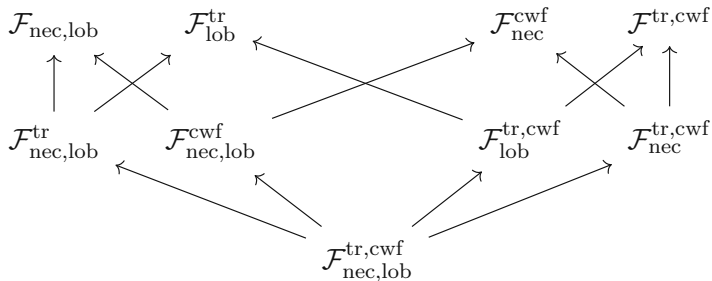
Completeness

Theorem

GL is complete for finite transitive irreflexive provability models having necessitation and Löb's rule.

Proof. Let $\text{GL} \not\vdash \phi$, so there is a finite transitive irreflexive Kripke model such that $\mathcal{K}, w \not\vdash \phi$. Then, we can transform this into a finite transitive irreflexive Kripke model with necessitation $\mathcal{P}$ by defining $\text{Ax}(\mathcal{L}_w) = \{\phi \mid \mathcal{K}, w \vdash \Box \phi\}$ and $\text{Rul}(\mathcal{L}_w) = \{\text{mp}, \text{nec}\}$. All left to show is that it has Löb's rule, so assume that for $v \sqsubset w$ we have that $\vdash_v \Box \phi \rightarrow \phi$. This means that $\mathcal{K}, v \vdash \Box(\Box \phi \rightarrow \phi)$ and (by Löb's axiom) $\mathcal{K}, v \vdash \Box \phi$. Since $\mathcal{K}, v \vdash \Box \phi \rightarrow \phi$ we obtain $\mathcal{K}, v \vdash \phi$. $\square$

Many soundness and completeness



And the same with finite frames.

Bifiniteness

Is it possible to easily construct provability models without Kripke frames? Also, can we find some class of models such that $\mathcal{P}, w \models \phi$ is decidable? **Yes**.

Definition

A provability premodel is called *bifinite* if its sets of worlds is finite, the set of axioms of each of its logics is finite and the set of rules of each of its logics is empty.

The cwf tree provability premodels can be extended to provability models uniquely up to l-isomorphism. Additionally in the bifinite case, using the original premodel, we can construct an algorithm to check the satisfiability of a formula in the extension.

Finitary models

Definition

A model is *finitary* if it is l-isomorphic to a provability model obtained from the minimum extension of a bifinite cwf tree provability model.

Finitary models are easy to construct and, if we keep the original premodel at hand, we can decide satisfiability in any of its worlds. Then, the following theorem shows that provability models is a satisfactory semantics for GL

Theorem

GL is complete for the class of finitary models.

Interpretability Logic

Let us change the language to

$$\phi ::= p \mid \perp \mid \phi \rightarrow \phi \mid \phi \triangleright \phi.$$

We will understand $\phi \triangleright \psi$ as $\text{PA} + \phi$ interprets $\text{PA} + \psi$. Interpretations are not easy to define and we do not need to know its precise meaning to understand the rest of the slides. You can understand $\phi \triangleright \psi$ as every model of $\text{PA} + \phi$ can be transformed into a model of $\text{PA} + \psi$. Note that under this view we can define

$$\Box \phi := \neg \phi \triangleright \perp, \quad \Diamond \phi := \neg(\phi \triangleright \perp).$$

The logic ILM

Definition. ILM is the logic whose only rule is mp and whose axioms are

1. ϕ for any propositional tautology and axioms (K), (4) and (L) in $\mathcal{L}_{\triangleright}$;
2. $\Box(\phi \rightarrow \psi) \rightarrow \phi \triangleright \psi$ for $\phi, \psi \in \mathcal{L}_{\triangleright}$;
3. $(\phi \triangleright \psi) \rightarrow (\psi \triangleright \chi) \rightarrow (\phi \triangleright \chi)$ for $\phi, \psi, \chi \in \mathcal{L}_{\triangleright}$;
4. $(\phi \triangleright \chi) \rightarrow (\psi \triangleright \chi) \rightarrow (\phi \vee \psi \triangleright \chi)$ for $\phi, \psi, \chi \in \mathcal{L}_{\triangleright}$;
5. $\Diamond \phi \triangleright \phi$ for $\phi \in \mathcal{L}_{\triangleright}$;
6. $\phi \triangleright \psi \rightarrow (\Box \chi \wedge \phi) \triangleright (\Box \chi \wedge \psi)$ for $\phi, \psi, \chi \in \mathcal{L}_{\triangleright}$;
7. $\Box \phi$ where ϕ is one of the axioms above.

Provability models for interpretability

In PA we have the following equivalence:

$$\text{PA} + \phi \text{ interprets } \text{PA} + \psi \text{ iff } \Pi_1\text{-Con}(\text{PA} + \psi) \subseteq \Pi_1\text{-Cons}(\text{PA} + \phi).$$

As $\Box \phi$ are Σ_1 -sentences, we can understand that in the modal language Π_1 -sentences are the ones of shape $\Diamond \phi$. Then, we define

$$\mathcal{P}, w \models \phi \triangleright \psi \text{ if } \forall v \Box w \forall \chi'. \vdash_u \psi \rightarrow \Diamond \chi \text{ implies } \vdash_u \phi \rightarrow \Diamond \chi'.$$

Soundness will hold for models that are (cwf or have Löb's rule) and have necessitation. Soundness of axioms (2), (3), (4) are straightforward.

Soundness of other axioms

Axiom $\Diamond \phi \triangleright \phi$. We want to show that $\mathcal{P}, w \models \Diamond \phi \triangleright \phi$, so assume that $v \sqsupset w$ and $\vdash_v \phi \rightarrow \Diamond \chi'$, i.e., $\vdash_v \Box \neg \chi' \rightarrow \neg \phi$. Using closure under necessitation (and since K is contained in L_v) we have that $\vdash_v \Box \Box \neg \chi' \rightarrow \Box \neg \phi$. Since $\vdash_v \Box \neg \chi' \rightarrow \Box \Box \neg \chi'$ (modal completeness) we obtain that $\vdash_v \Box \neg \chi' \rightarrow \Box \neg \phi$, i.e., $\vdash_v \Diamond \phi \rightarrow \Diamond \chi'$.

Montagna's. Assume $\mathcal{P}, w \models \phi \triangleright \psi$ and let $v \sqsupset w$ such that $\vdash_v \Box \chi \wedge \psi \rightarrow \Diamond \chi'$. Then (as L_v contains logic K by modal completeness for axiom K) $\vdash_v \psi \rightarrow \Diamond \neg \chi \vee \Diamond \chi'$, so $\vdash_v \psi \rightarrow \Diamond (\neg \chi \vee \chi')$. Thanks to $\mathcal{P}, w \models \phi \triangleright \psi$ we have that $\vdash_v \phi \rightarrow \Diamond (\neg \chi \vee \chi')$, i.e., $\vdash_v \Box \chi \wedge \phi \rightarrow \Diamond \chi'$.

Completeness

Completeness can be established using Veltman's semantics.

In fact, it is possible to show completeness for finitary provability models.

Future work

1. Show arithmetical completeness using directly provability models (it is not what occurs in Mojtaba's work on intuitionistic provability logic).
2. Define provability models for
 - 2.1 ILP, ER, and in general all provability logics.
 - 2.2 Intuitionistic modal logic with $\Box$ and $\Diamond$: IGL.
 - 2.3 The logic of proofs.
3. In the paper we also define provability models for GLP. However, we have not found (yet) a natural class of provability models for GLP(similar to finitary models in GL and ILM).

04

Answering Q2: proofs

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Q2 Can we find a more uniform approach to the first step of Solovay's proof?

The canonical model

Definition

Given a (consistent) modal logic L define the Kripke model $\mathcal{K}_L$ as

1. The worlds are the maximal consistent L -theories.
2. We have $\Gamma \sqsubset \Delta$ iff $\{\phi \mid \Box \phi \in \Gamma\} \subseteq \Delta$.
3. For $V(p) = \{\Gamma \mid p \in \Gamma\}$.

It is easy to show by induction on ϕ that for any Γ , we have $\mathcal{K}_L, \Gamma \models \phi$ iff $\phi \in \Gamma$. This means that $\mathcal{K}_L \models \phi$ iff $L \vdash \phi$, i.e., L is sound and complete for the class of models $\{\mathcal{K}_{GL}\}$.

Lesson learned

The previous slide means that what is hard is not to find a class of models such that $\mathcal{M}$ is sound and complete for L . What is hard is to find one which **can be easily embedded into arithmetic**.

For GL is unproblematic: we can choose finite irreflexive transitive trees. For the provability logic of HA the interaction between intuitionistic and modal relation makes it harder. For GLP (if we want to look to a class of frames) it is impossible. However, the proof theory for provability have been developed and simplified these last years (Shamkanov; 2014), (Savateev and Shamkanov; 2021), (Goré, Ramanayake and Shilito; 2021) and (SM, Studer and Zenger; 2024).
What if we put it to use?

The intuition

We have two intuitions.

1. For the proof theoretic minded: for Solovay's proof we just need a **witness** of the unprovability of a formula that can be embedded into arithmetic. Not having a proof in a sequent calculus is a Π_1 -statement, but if the calculus is well-behaved and with the help of combinatorial game theory we can obtain a Σ_1 -statement equivalent to it. Here we have our witness, semantic free.
2. For the semantic minded: that there is no nice class of frames does not imply the inexistence of a nice class of **models** which is complete for the logic. The description of this class of models, which are going to be finite trees and thus easy to embed into arithmetic, comes from a sequent calculus.

The details, *grosso modo*

1. Take your favorite sequent calculus for the logic. Define the proof search game and show that it is (positionally) determined and that winning strategies of verifier correspond to proofs.
2. Then, unprovability of a formula (or sequent) correspond to a winning strategy of refuter.
3. Assume ϕ is unprovable, and obtain a refutation (winning strategy for refuter). This gives a tree by considering the possible games which are consistent with the given refutation.
4. Look at the positions of verifier in that tree, the sequents in those positions are saturated. Embed the tree of the verifier positions in arithmetic, being 'true' in a sequent will correspond to belonging to the antecedent of the sequent.

Future work

With this we (re)proved arithmetical completeness of GL and of other Kripke incomplete logics such as ER and GLP.

1. Adapt the method to other provability logics: specially interesting the intuitionistic case.
2. Can we obtain an alternative description of the class of models that we obtain through proof search? Can we obtain a *semantic* description of them?
3. Develop the proof theory of provability logics. Can we adapt this method if other kind of sequent calculus is used?

06

Conclusions and bibliography

Conclusions

1. Provability models are the initial steps towards a solution of **Q1**. They also appear naturally in Mojtaba's previous work and should also provide an answer to **Q2** (WIP). Still, it may still be possible to refine some notions (getting a notion of frame, obtaining better correspondence between modal axioms and properties of the models, ...).
2. Sequent calculus provides a good answer to **Q2**, we have already some non-trivial examples (ER and GLP). Still, we have to explore more logics to be sure. In any case, this shows that (nowadays) syntactic methods can reach the level of semantic methods for provability logic and that they provide an useful tool to have in the provability logician toolkit.

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A1

Arith. compl. of ER via proofs

ER Logic

We use the language

$$\phi ::= p \mid \perp \mid \phi \rightarrow \phi \mid \Box_0 \phi \mid \Box_1 \phi.$$

Then ER is the set of formulas closed under mp and nec for both modalities and contains axioms

1. Propositional tautologies in the language $\mathcal{L}_2$.
2. $\Box_i(\phi \rightarrow \psi) \rightarrow \Box_i \phi \rightarrow \Box_i \psi$ for $i \in \{0, 1\}$.
3. $\Box_i(\Box_i \phi \rightarrow \phi) \rightarrow \Box_i \phi$ for $i \in \{0, 1\}$.
4. $\Box_i \phi \rightarrow \Box_{1-i} \Box_i \phi$ for $i \in \{0, 1\}$.
5. $\Box_1(\Box_0 \phi \rightarrow \phi)$.

ER Sequent calculus

Calculus from (SM and Studer; 2026). We have two kind of sequents $\Gamma \Rightarrow_i \Delta$ interpreted as

$$(\Gamma \Rightarrow_0 \Delta)^b := \bigwedge \Gamma \rightarrow \bigvee \Delta, \quad (\Gamma \Rightarrow_1 \Delta)^b := \Box_1 \left(\bigwedge \Gamma \rightarrow \bigvee \Delta \right).$$

And rules

$$\begin{array}{c} \frac{}{p, \Gamma \Rightarrow_i p, \Delta} \text{ax} \quad \frac{}{\perp, \Gamma \Rightarrow_i \Delta} \perp\text{L} \quad \frac{\Gamma \Rightarrow_i \Delta}{\Gamma \Rightarrow_i \perp, \Delta} \perp\text{R} \\[10pt] \frac{\psi, \Gamma \Rightarrow_i \Delta \quad \Gamma \Rightarrow_i \phi, \Delta}{\phi \rightarrow \psi, \Gamma \Rightarrow_i \Delta} \rightarrow\text{L} \quad \frac{\phi, \Gamma \Rightarrow_i \psi, \Delta}{\Gamma \Rightarrow_i \phi \rightarrow \psi, \Delta} \rightarrow\text{R} \\[10pt] \frac{\Box_i \Sigma_i, \Box_{1-i} \Sigma_{1-i}, \Box_i \phi \Rightarrow_i \phi}{\Box_i \Sigma_i, \Box_{1-i} \Sigma_{1-i}, \Gamma \Rightarrow_j \Box_i \phi, \Delta} \Box_i \quad \frac{\Box_0 \phi, \Gamma \Rightarrow_1 \Delta}{\Box_0 \phi, \Gamma \Rightarrow_1 \Delta} \text{ER} \end{array}$$

And we can define *saturation*.

Properties of Refutations

We have a refutation whose nodes are $W = \{0, \dots, N\}$. W_i are the non-zero nodes with a $\Rightarrow_i$ -sequent.

Lemma

We have the following for $x \in W \setminus \{0\}$ ($\Gamma_x \Rightarrow_{i_x} \Delta_x$ is the sequent at node x):

1. $\perp \notin \Gamma_x$ and $\Gamma_x \cap \Delta_x = \emptyset$. Additionally, $\Gamma_x \Rightarrow_{i_x} \Delta_x$ is saturated.
2. $\Box_i \psi \in \Delta_x$ implies there is a y such that $x \sqsupset y \in W_i$ and $\psi \in \Delta_y$.
3. If $\Box_i \psi \in \Gamma_x$ and $x \sqsupset^+ y$ then $\Box_i \psi \in \Gamma_i$. If $y \in W_i$ then $\psi \in \Gamma_y$.
4. If $\Box_0 \psi \in \Gamma_x$ and $x \in W_1$ then $\psi \in \Gamma_x$.

Solovay sentences

Using the refutation and Solovay's ideas we can define arithmetical sentences

$(\lambda_x)_{x \in W}$ such that

S1 $EA \vdash \bigvee_{x \in W} \lambda_x$.

S2 $EA \vdash \neg(\lambda_x \wedge \lambda_y)$ for $x \neq y$.

S3 $EA \vdash \lambda_x \rightarrow \blacklozenge_i \lambda_y$ for $x \sqsubset y \in W_i$.

S4 $EA \vdash \lambda_x \rightarrow \blacksquare_i \bigvee_{y|x \sqsubseteq +} \lambda_y$.

S5 $EA \vdash \lambda_x \rightarrow \blacksquare_i \neg \lambda_x$ for $x \in W_i$.

S6 $\mathbb{N} \vdash \lambda_0$.

Lemma

$PA \vdash \neg \lambda_x$ for $x \in W_0$.

We define the arithmetical interpretation

$$p^* = \bigvee_{x|p \in \Gamma_x} \lambda_x.$$

Truth Lemma

Lemma

For any $x \in W \setminus \{0\}$, we have that

$\psi \in \Gamma_x$ implies $\text{EA} \vdash \lambda_x \rightarrow \psi^*$, $\psi \in \Delta_x$ implies $\text{EA} \vdash \lambda_x \rightarrow \neg\psi^*$.

Completeness

Assume $ER \not\vdash \phi$, so there is a refutation ending in $\Rightarrow_0 \phi$. To get a contradiction assume that $EA \vdash \phi^*$. Then we know that $1 \in W_0$ must be reached using only propositional rules, so $\psi \in \Delta_1$. By the truth lemma we get that $EA \vdash \lambda_1 \rightarrow \neg\phi^*$, so $EA \vdash \neg\lambda_1$. Using necessitation we get that $EA \vdash \Box_0 \neg\lambda_1$ and by S3 we obtain that $EA \vdash \neg\lambda_0$. Soundness of EA gives $\mathbb{N} \models \neg\lambda_0$ and with S6 we obtained the desired contradiction.

So completeness has been proven.

Thank you.